LESSON 6.7
USING THREE-COLUMN TABLES OF EQUIVALENT RATIOS

LESSON INTRODUCTION

MA.6.AR.3.3 Extend previous understanding of fractions and numerical patterns to generate or complete a two- or three-column table to display equivalent part-to-part ratios and part-to-part-to-whole ratios.

MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates, and unit rates, including comparisons, mixtures, ratios of lengths and conversions within the same measurement system.

LEARNING TARGETS

- I can use three-column tables to display equivalent ratios.
- I can solve problems using three-column tables of equivalent ratios.

BENCHMARK COHERENCE

BUILDING ON

MA.5.AR.3.1 Given a numerical pattern, identify and write a rule that can describe the pattern as an expression.

WORKING TOWARD

MA.7.AR.3.2 Apply previous understanding of ratios to solve real-world problems involving proportions.

ESSENTIAL QUESTIONS

- How do you display equivalent ratios in a three-column table?
- How do you solve problems using three-column tables of equivalent ratios?

LESSON NARRATIVE

Students learn how to use three-column ratio tables to display equivalent ratios and solve problems. Then, students practice using three-column ratio tables to display equivalent ratios and solve problems. Finally, students write a short note to a friend who missed class that summarizes today's lesson.

COMPONENT SUMMARY

6.7.1 WARM-UP *(~5 MIN.)*

Compare the brain storage capacity of cats and humans

6.7.2 GUIDED INSTRUCTION *(20-30 MIN.)*

Use three-column ratio tables to display equivalent ratios and solve problems

6.7.3 GUIDED PRACTICE *(10-15 MIN.)*

Students practice using three-column ratio tables to display equivalent ratios and solve problems.

6.7.4 YOUR TURN! *(10 MIN.)*

Use three-column ratio tables to display equivalent ratios and solve problems

6.7.5 WRAP-UP *(~5 MIN.)*

Students write a short note to a friend who missed class that summarizes today's lesson

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- Ratio

MATERIALS

- (Optional) Note cards
- (Optional) Chart paper
- (Optional) Markers

ADDITIONAL PREPARATION

N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may confuse part-to-part and part-to-whole ratios.
- Students may multiply when they should be dividing or divide when they should be multiplying (i.e., students may multiply by 3 in 6.7.2 Guided Instruction question 3a rather than dividing by 3 or multiplying by $\frac{1}{3}$).

THINGS TO CONSIDER

- Have students draw an arrow from each row of the table with the number by which they need to multiply or divide the numbers in the row to maintain an equivalent ratio. Also, have students write their calculations out so they can check their work. This will help prevent errors.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

Consider having students engage in 6.7.1 Warm-Up using the Think-Pair-Share routine. Initially, have students independently solve the problems. Then, ask students to discuss how they solved the problems and the answer they obtained with a partner. Have students resolve any differences in their answers through discussion. The discussion will help develop student understanding and confidence.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

This lesson presents a multitude of opportunities for rich dialogue on

real-world applications. The applications in this lesson include brain capacity (6.7.1 Warm-Up), basketball, mixing concrete (6.7.2 Guided Instruction), mixing paint, walking (6.7.3 Guided Practice), and mixing metals (6.7.4 Your Turn!). Students use three-column ratio tables to display equivalent ratios and solve problems in these contexts.

DIFFERENTIATE INSTRUCTION

Have students draw an arrow from each row of the table with the number by which they need to multiply or divide the numbers in the row to maintain an equivalent ratio.

For example, notice the table of values in 6.7.2 Guided Instruction. Students may apply rules of distribution, thinking similar strategies would apply. Consider proving both visual and verbal entry points for students to make sense of which values should be multiplied and the reasoning behind the operations.

Cement	Sand	Gravel
1	2	3
$\frac{1}{3}$	$\frac{2}{3}$	1
6.5	13	19.5
5	10	15

6.7.1 WARM-UP: BRAIN CAPACITY

Component Overview: *Students compare the brain storage capacity of cats and humans. Consider engaging students in a Think-Pair-Share as they answer questions 1 and 2.*

TEACHER MOVES

Think-Pair-Share

Initially, have students independently solve the problems. Then, have students discuss how they solved the problems and the answers they obtained with a partner. Have students resolve any differences in their answers through discussion. The conversation may illuminate more effective strategies for problem-solving.

6.7.1 Warm-Up: Brain Capacity

A human brain can store about 3,500 trillion bytes of data, while a cat's brain can store about 98 trillion bytes of data.

1. **What is the brain storage capacity of human to cat?**
   3,500 trillion bytes to 98 trillion bytes or about 35.7 bytes to 1 byte.

2. **About how many brains of cats would it take to store the same amount of data as the human brain?**
   It would take approximately 36 cats to store the same amount of data as the human brain.

6.7.2 Guided Instruction: Using Three-Column Ratio Tables

Ratios can be used to make part-to-part or part-to-whole comparisons. When comparing parts of a mixture to the whole, three-column ratio tables can be useful to represent the relationships between quantities.

1. Kiara is practicing her free-throw shot at basketball practice. She averages 4 made shots for every 2 missed shots. The table shown can be used to represent this situation.

Number of Shots Made	Number of Shots Missed	Total Shots Taken
4	2	6
16	8	24
2	1	3

   a. When determining this rate during practice, Kiara missed 8 free-throw shots. Complete the second row of the table to show how many shots she made and how many total shots she took.

   b. How was each value in the second row determined?

   We are told that she missed 8 shots. Previously she missed 2 shots. Dividing 8 by 2 we obtain 4. This tells us that each value in row one should be multiplied by 4. Multiplying 4 by 4 we see she made 16 shots. Multiplying 6 by 4 we see she took a total of 24 shots. Checking our work, the shots made and shots missed equals the total shots.

6.7.2 GUIDED INSTRUCTION: USING THREE-COLUMN RATIO TABLES

Component Overview: *Students use a table to find and represent the number of free-throw shots Kiara took at basketball practice. Students find missing values and then explain how they used the table to do so. Students then apply this to a table representing the concrete mixture, using information in the table to find other missing information.*

DIFFERENTIATE INSTRUCTION

Notice the opportunities for verbal mathematical reasoning in this component. Some students may benefit from verbalizing their responses (questions 1b and 2a) before putting them in writing. Ensure that students have an opportunity to engage in discussion to provide this entry point for learners.

TEACHER MOVES

Turn and Talk

To provide an opportunity for student discussion, engage the class in a “Turn and Talk” routine before each written response question. Read the question aloud for the class, then have students share their responses with their partners before writing. As students are discussing, circulate to collect informal data on how students are determining the values in each row. Consider bringing the class back as a group and having students share particularly efficient strategies or draw attention to a common misconception.

6.7.2 GUIDED INSTRUCTION: USING THREE-COLUMN RATIO TABLES (CONTINUED)

ENRICHMENT

To encourage mathematical reasoning as students fill in the tables, ask questions like:

- Do the answers in your tables make sense? How do you know?
- How did you use the table to help determine what should go in the next row?
- How can you check your work to ensure you filled in the table correctly?

QUESTIONING

Challenge advanced learners to write their own real-world problems that can be displayed and solved using a three-column ratio table. Students can solve their own problems or swap with a partner and solve each other's problems.

2. Kiara had the opportunity to take 3 free-throw shots in her most recent game, and her accuracy remained the same as in practice. Complete the third row of the table to show how many free-throw shots she made and missed in the game.

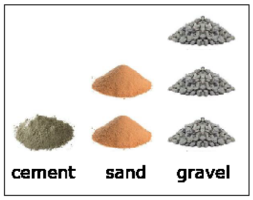
   a. How was each value in the third row determined?
   We are told that she took 3 shots. Previously she took 6 shots. Dividing 6 by 3 we obtain 2. This tells us that each value in row one should be divided by 2. Dividing 4 by 2 we see she made 2 shots. Dividing 2 by 2 we see she missed a total of 1 shot. Checking our work, the shots made and shots missed equals the total shots.

   b. Complete the statement.

   For every 3 free-throw shots taken, Kiara makes __2__ shot(s) and misses __1__ shot(s) on average. The ratio of missed shots to made shots to total shots is __1__ : __2__ : __3__.

3. Three-column tables can also be useful when comparing mixtures that contain more than two parts.

   Concrete can be created by mixing 1 part cement with 2 parts sand and 3 parts gravel. These dry ingredients are first combined before being mixed with water.



   a. Complete the table to determine equivalent mixtures of concrete. Show how the values in each row are determined using arrows and by indicating the factor used.

Divide by 3
Multiply by 6.5
Multiply by 5

Cement	Sand	Gravel
1	2	3
$\frac{1}{3}$	$\frac{2}{3}$	1
6.5	13	19.5
5	10	15

   b. Marc uses one-gallon buckets to measure out the quantities needed in his concrete mixture. How many gallons of dry mix will he have in one batch of concrete?
   Marc will need 6 gallons of dry mix to make one batch of concrete.

6.7.3 Guided Practice: Using Three-Column Ratio Tables

1. A particular shade of maroon paint calls for 5 cups of red paint to be mixed with 3 cups of blue paint. Complete the table to find the quantities needed for different batches of paint.

Red Paint (cups)	Blue Paint (cups)	Maroon Paint (cups)
5	3	8
20	12	32
1.25	0.75	2
2.5	1.5	4
50	30	80

a. If the mixture in the first row represents one batch of maroon paint, how many batches are in 80 cups of maroon paint?
Ten batches are in 80 cups of maroon paint.

b. Explain how this relates to the factor used to complete the last row in the table.
Each number of cups in the first row was multiplied by 10 to determine the number of cups in the last row. Therefore, if the first row represents one batch, the last row will represent 10 batches.

2. There are 16 cups in one gallon. Use this information and the table below to determine how many gallons of paint the 80-cup mixture of maroon paint makes.

Cups	Gallons
16	1
80	5

3. In April, Kyla went for a walk on 4 days for every 1 day that she did not walk. There are 30 days in April. Use this information to complete the ratio table.

Walking Days	4	24
Non-Walking Days	1	6
Total Days	5	30

Complete the statement.

Throughout the year, Kyla continued walking 4 days for every 1 day that she did not walk. Based on this, in November, Kyla walked 24 of the 30 days.

6.7.3 GUIDED PRACTICE: USING THREE-COLUMN RATIO TABLES

Component Overview: *Students begin this Guided Practice by using a three-column table to find the quantities needed for different paint mixtures, then explain how they completed the table. Students then complete two more tables based on given real-world contexts, applying their knowledge of ratios and unit rates to find unknowns in the tables.*

DIFFERENTIATE INSTRUCTION

To provide a visual entry point, have students draw an arrow from each row of the table with the number by which they need to multiply or divide the numbers in the row to maintain an equivalent ratio (see example in 6.7.4 Your Turn! answer key).

TEACHER MOVES

After students complete the tables in questions 1, 2 and 3 consider having students share their strategies in small groups. This will provide an opportunity for students to verbalize their thinking and compare their strategies to others' means of solving, allowing them to make connections and/or determine more efficient means of solving.

6.7.4 YOUR TURN!

Component Overview: *Students continue practicing using three-column ratio tables to display equivalent ratios and solve problems.*

ENRICHMENT

Students may struggle to understand how to begin filling in the table. Consider scaffolding this component by asking questions like:

- What do you need to determine first in order to find the rest of the values in the table?
- How can you determine the weight of bronze using the information in the table? Does it make sense to multiply or add the weights of copper and tin to find the weight of bronze?

6.7.4 Your Turn!

1. **Bronze is a type of metal used to make many products such as ship propellers and parts of musical instruments. Bronze is made by mixing copper and tin in a ratio of 8.8 to 1.2. Complete the table where each quantity is measured in kilograms (kg). Then, answer the questions that follow.**

Copper (kg)	8.8	22	$7\frac{1}{3}$
Tin (kg)	1.2	3	1
Bronze (kg)	10	25	$8\frac{1}{3}$

   a. **How many kilograms of tin are needed to make 25 kilograms of bronze?**
   3 kilograms of tin are needed to make 25 kilograms of bronze.

   b. **To make bronze, how many kilograms of copper need to be melted per melted kilogram of tin?**
   $7\frac{1}{3}$ kilograms of copper are needed per kilogram of tin to make bronze.

   c. **Draw an arrow to connect each question above to where the necessary information to answer can be found in the ratio table.**

6.7.5 Wrap-Up: Cool Down

1. Write a short note to a friend who missed class that summarizes today's lesson.

Answers will vary. Today we learned how we can use three-column tables to display equivalent ratios. The ratio in each row is the same. Each row is also a multiple of another row in the table. We used these facts to solve problems using three-column tables of equivalent ratios. The table helps us keep the ratios organized and determine missing values in the table.

6.7.5 WRAP-UP: COOL DOWN

Component Overview: *Students write a short note to a friend who missed class that summarizes today's lesson.*

DIFFERENTIATE INSTRUCTION

Consider how to utilize the notes written by students.

- Provide as notes for absent students
- Keep as personal anchor charts for students to review or study
- Trade notes with a partner, and your partner for homework must provide an example of demonstration based on the note written.